

Given: Due:	Further Maths (Core Pure) HW 08	Name:
Chapter 6 (6.5 & 6.6): Matrices Simultaneous equations & 3x3 inverses		
1)		
(i) Find the determinant of the matrix $\mathbf{M} = \begin{pmatrix} k & 2 & 3 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix}$ in terms of k .		[3]
(ii) State the value of k for which $\mathbf{M}$ is singular.		[1]

(i) $\begin{vmatrix} k & 2 & 3 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{vmatrix} = k(-2 + 1) - 2(-3 + 2) + 3(3 - 4)$

$$= -k + 2 - 3$$

$$= -k - 1$$

(ii) $k = -1$

[3]

[1]

2)		
$\mathbf{M} = \begin{pmatrix} 2 & 1 & -3 \\ 4 & -2 & 1 \\ 3 & 5 & -2 \end{pmatrix}$		
(a) Find $\mathbf{M}^{-1}$ giving each element in exact form.		(2)
(b) Solve the simultaneous equations		
$2x + y - 3z = -4$		
$4x - 2y + z = 9$		
$3x + 5y - 2z = 5$		
		(2)

This question was generally well done with the majority gaining full marks for (a) and (b).

(a) There were many correct responses many with clear evidence of manual calculation in some cases, although it was expected that students used their calculators. A few wrote each term as a decimal so were unable to gain more than one mark as an exact form was required. Many fully correct answers were seen.

(b) The majority of students used their inverse correctly to find the coordinates using a calculator. Some students chose a manual simultaneous equation method of solution, but this often led to errors. Others seem to have used a calculator to solve the simultaneous equations rather than take the inverse matrix approach.

Question	Scheme EXAM PAPERS PRACTICE	Marks	AOs
(a)	$\mathbf{M}^{-1} = \frac{1}{69} \begin{pmatrix} 1 & 13 & 5 \\ -11 & -5 & 14 \\ -26 & 7 & 8 \end{pmatrix}$	B1 B1	1.1b 1.1b
		(2)	
(b)	$\frac{1}{69} \begin{pmatrix} 1 & 13 & 5 \\ -11 & -5 & 14 \\ -26 & 7 & 8 \end{pmatrix} \begin{pmatrix} -4 \\ 9 \\ 5 \end{pmatrix} = \dots$	M1	1.1b
	$x = 2, y = 1, z = 3 \text{ or } (2, 1, 3) \text{ or } 2\mathbf{i} + \mathbf{j} + 3\mathbf{k} \text{ or } \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$	A1	1.1b
		(2)	
(c)	The point where three planes meet	B1ft	2.2a
		(1)	
(5 marks)			
Notes			
(a) B1: Evidence that the determinant is ± 69 (may be implied by their matrix e.g. where entries are not in exact form: $\pm \begin{pmatrix} 0.014 & 0.188 & 0.072 \\ -0.159 & -0.072 & 0.203 \\ -0.377 & 0.101 & 0.116 \end{pmatrix}$) (Should be mostly correct) Must be seen in part (a). B1: Fully correct inverse with all elements in <u>exact</u> form (b) M1: Any complete method to find the values of x, y and z (Must be using their inverse if using the method in the main scheme) A1: Correct coordinates A solution not using the inverse requires a complete method to find values for x, y and z for the method mark. Correct coordinates only scores both marks. (c) B1: Describes the correct geometrical configuration. Must include the two ideas of planes and meet in a point with no contradictory statements. This is dependent on having obtained a unique point in part (b)			

3)

$$\mathbf{A} = \begin{pmatrix} -2 & 1 & -3 \\ k & 1 & 3 \\ 2 & -1 & k \end{pmatrix}, \text{ where } k \text{ is a constant}$$

Given that the matrix $\mathbf{A}$ is singular, find the possible values of k .

(4)

The overwhelming majority of students knew that they had to solve the equation $\det \mathbf{A} = 0$, with only a very small number unaware of the definition of "singular". Most students processed the determinant conventionally, usually by using row 1, although the Rule of Sarrus was seen quite regularly. Sign and algebraic errors were rarely seen and most students obtained the correct values for k and the full four marks.

Question Number	Scheme	EXAM PAPERS PRACTICE Notes	Marks
	$A = \begin{pmatrix} -2 & 1 & -3 \\ k & 1 & 3 \\ 2 & -1 & k \end{pmatrix}$		
	$\det A = -2(k+3) - (k^2 - 6) - 3(-k-2)$ row1 or e.g. $\det A = -k(k-3) + (-2k+6) - 3(2-2)$ row2 $\det A = 2(3+3) + (-6+3k) + k(-2-k)$ row3 $\det A = -2(k+3) - k(k-3) + 2(3+3)$ col1 $\det A = -(k^2 - 6) + (-2k+6) + (-6+3k)$ col2 $\det A = -3(-2-k) - 3(2-2) + k(-2-k)$ col3	M1: Correct attempt at determinant (3 'elements' (may be implied if one is zero) with at least two elements correct). Note that there are various alternatives depending on the choice of row or column. A1: Correct determinant in any form	M1A1
	Note that e.g. $\det A = -2 \begin{vmatrix} 1 & 3 \\ -1 & k \end{vmatrix} - \begin{vmatrix} k & 3 \\ 2 & k \end{vmatrix} - 3 \begin{vmatrix} k & 1 \\ 2 & -1 \end{vmatrix}$ scores no marks until the determinants are 'extracted'.		
	$-2(k+3) - (k^2 - 6) - 3(-k-2) = 0 \Rightarrow k = \dots$	Sets their $\det A = 0$ ($= 0$ may be implied) and attempts to solve a 3 term quadratic (see general guidance) as far as $k = \dots$ NB Correct quadratic is $k^2 - k - 6 = 0$	M1
	$(k+2)(k-3) = 0 \Rightarrow k = -2, 3$	Both values correct	A1
			(4)
			Total 4

4) The population of chimpanzees in a particular country consists of juveniles and adults. Juvenile chimpanzees do not reproduce.

In a study, the numbers of juvenile and adult chimpanzees were estimated at the start of each year. A model for the population satisfies the matrix system

$$\begin{pmatrix} J_{n+1} \\ A_{n+1} \end{pmatrix} = \begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix} \begin{pmatrix} J_n \\ A_n \end{pmatrix} \quad n = 0, 1, 2, \dots$$

where a is a constant, and J_n and A_n are the respective numbers of juvenile and adult chimpanzees n years after the start of the study.

(a) Interpret the meaning of the constant a in the context of the model.

(1)

At the start of the study, the total number of chimpanzees in the country was estimated to be 64 000

According to the model, after one year the number of juvenile chimpanzees is 15 360 and the number of adult chimpanzees is 43 008

(b) (i) Find, in terms of a

$$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1}$$

(3)

(ii) Hence, or otherwise, find the value of a .

(3)

(iii) Calculate the change in the number of juvenile chimpanzees in the first year of the study, according to this model.

(2)

Given that the number of juvenile chimpanzees is known to be in decline in the country,

(c) comment on the short-term suitability of this model.

(1)

A study of the population revealed that adult chimpanzees stop reproducing at the age of 40 years.

(d) Refine the matrix system for the model to reflect this information, giving a reason for your answer.

(There is no need to estimate any unknown values for the refined model, but any known values should be made clear.)



This question provided a very challenging conclusion to the paper, but the majority of candidates made some kind of attempt at parts (a) and (b)(i) at least. Many did not make further progress, however, possibly due to time constraints, or perhaps not knowing how to proceed. The modal mark of 3 was generally gained from a correct answer of (b)(i) and little else.

(a) A correct interpretation of the constant a was rarely seen with many candidates incorrectly believing it to be the number of surviving juveniles, rather than the number of juveniles who remained as juveniles in the following year.

(b)(i) As noted above, this part was answered well by the majority of candidates, with almost all able to find the determinant of the 2×2 matrix and most going on correctly to find its inverse. A common error was that only one sign change had been made, while some did not form the adjoint matrix at all, but in general 3 was the most common mark here.

(b)(ii) Attempts at this part and onwards were less common, but many did nevertheless make some kind of attempt. Of those who did proceed, the majority attempted to use the inverse matrix to determine the values of a and J_0 but had found it a challenge to match up the results of their matrix multiplication with the total of 64,000 chimpanzees. Algebra here, in finding a , was often resulted in error. Similar levels of success were achieved by those candidates who set up and solved associated simultaneous equations, and these more commonly achieved the values of J_0 correctly.

(b)(iii) Many candidates failed to see the connection between this part and (b)(iii) and did not find the value of J_0 unless they had already done so as a part of their method. Even in such cases they did not always go on to achieve the correct answer. But there were also many candidates who did make good attempts at this part, achieving the correct difference.

(c) Candidates who achieved a value for J_0 would generally go on to make a comment for this part, although not all were able to make the link correctly. There were a large number of responses that were not based on the solutions from part (b), but instead tried to make general comment about long term unsuitability of the model. Many such of these had no answer to part (b) and so could not access the mark in any case.

(d) Very few candidates even attempted this part. Amongst those who did, only a very small minority realised the need to extend from a 2×2 matrix system to a 3×3 system by introducing a third category of chimpanzee, and fully correct answers were very rare. Many focussed more on trying to adapt one entry in the given system, either just in its value, or by adding a variable based on the number of mature chimpanzees, while others attempted to adjust by subtraction of an extra vector term. However, none of these methods would enable the new number of mature chimpanzees to be determined and so no credit could be given for them.

Question	Scheme	Marks	AOs
(a)	a represents the proportion of juvenile chimpanzees that (survive and) remain juvenile chimpanzees the next year.	B1	3.4
		(1)	
(b)(i)	Determinant = $0.82a - 0.08 \times 0.15$	M1	1.1b
	$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1} = \dots \begin{pmatrix} 0.82 & -0.15 \\ -0.08 & a \end{pmatrix}$	M1	1.1b
	$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1} = \frac{1}{0.82a - 0.012} \begin{pmatrix} 0.82 & -0.15 \\ -0.08 & a \end{pmatrix}$	A1	1.1b
		(3)	
(ii)	$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1} \begin{pmatrix} 15360 \\ 43008 \end{pmatrix} = \frac{1}{0.82a - 0.012} \begin{pmatrix} 0.82 \times 15360 - 0.15 \times 43008 \\ (-0.08) \times 15360 + 43008a \end{pmatrix}$ OR forms equations $15360 = aJ_0 + 0.15 \times A_0$ $43008 = 0.08 \times J_0 + 0.82 \times A_0$	M1	3.1a
	$\frac{1}{0.82a - 0.012} [6144 + (43008a - 1228.8)] = 64000$ $\Rightarrow 4915.2 + 43008a = 64000(0.82a - 0.012) \Rightarrow a = \dots$ OR $A_0 = 64000 - J_0 \Rightarrow 43008 = 0.08 \times J_0 + 0.82 \times (64000 - J_0) = J_0 = \dots$ $\Rightarrow a = \frac{15360 - (64000 - J_0)}{J_0} = \dots$	M1	3.1a
	$a = \frac{5683.2}{9472} = 0.60$	A1	1.1b
		(3)	
(iii)	Initial juvenile population = $\frac{"6144"}{"0.48"} = 12800$	M1	3.4
	So change of 2560 juvenile chimpanzees	A1	1.1b
		(2)	
(c)	As the number of juveniles has increased, the model is not initially predicting a decline, so is not suitable in the short term. (Follow through their answer to (b) – but they must have made an attempt at it to find at least a value for J_0)	B1ft	3.5a
		(1)	
(d)	Third category needs to be introduced for chimpanzees aged 40 and above, mature chimpanzees M_n , and a matrix multiplication of increased dimension set up. Accept 3×3 , 3×2 or 2×3 matrices including all three categories in the column vector. The corresponding matrix model will have the form $\begin{pmatrix} J_{n+1} \\ A_{n+1} \\ M_{n+1} \end{pmatrix} = \begin{pmatrix} a & b & \underline{0} \\ 0.08 & c & 0 \\ 0 & d & e \end{pmatrix} \begin{pmatrix} J_n \\ A_n \\ M_n \end{pmatrix}$ (The underlined zero must be correct but do not be concerned about any values used in the other entries.)	M1	3.5c
		A1	3.3
		(2)	
(12 marks)			

Notes

(a)	B1	Correct interpretation. Need not mention survival but must be clear it is the (proportion of) juveniles that remain as juveniles the next year (ie those that survive but don't progress to adulthood). E.g. accept "(number of) juveniles who do not become adults" but do not accept "surviving juveniles".
(b)(i)	M1	Mark part (b) as a whole. Attempts the determinant in terms of a . Allow miscopies for the attempt. Allow $0.82a - 0.12$ as a slip.
	M1	Attempts the form of the inverse, swapped leading diagonals and sign changed on both off diagonals. Allow miscopies of the numbers but the signs must be correct.
	A1	Correct inverse matrix
(ii)	M1	Use the inverse matrix and attempts to find the initial juvenile and adult populations. (May have determinant 1 for this mark.) Alternatively, sets up simultaneous equations from the original system. $15360 = aJ_0 + 0.15 \times A_0$ and $43008 = 0.08 \times J_0 + 0.82 \times A_0$. Accept with J_n and A_n or other appropriate variables.
	M1	Uses the sum of initial populations equals 64000 in an attempt to find a . (May have determinant 1 for this mark.) If using alternative, use of e.g. $A_0 = 64000 - J_0$ in second equation to find J_0 , followed by attempt to find a . Award for an attempt to solve the equations, but don't be too concerned with the algebraic process as long as they are attempting to use all three equations.
	A1	Correct value, $a = 0.6$ (or 0.60 or $\frac{3}{5}$).
(iii)	M1	Uses their a to find the value of J_0 . This mark may be gained for work done in (ii) if the alternative has been used but must have come from a correct method.
	A1	Correct difference found, as long as there is no contradictory statement – so "decrease of 2560" is A0.
(c)	B1ft	Comments that the change is an increase so does not fit the model. Follow through their answer to (b) as long as at least a value for J_0 has been found. If a decrease has been found allow for commenting the model is suitable. If an answer is given to (b)(iii), follow through on whatever their answer is. If no answer has been given, but an initial population found, a comparison should be made between this value and 153600 with conclusion must be consistent with their answer for J_0
(d)	M1	Introduces a third category (may be <i>Mature</i> , <i>Elderly</i> or any suitable letter used) and sets up a matrix multiplication (the left hand side may be missing for this mark) with all three categories in the column vector. The dimension of the matrix should be 3 in at least either row or column, and there should be a 3×1 vector.
	A1	Sets up the new matrix equation, including both sides and making clear the zero (underlined) so that the correct progression that no new juveniles arise from the mature chimpanzees is clear. Overlook other values, though ideally the other two zeroes are shown too, to indicate mature chimpanzees do not regress to adulthood, and juveniles cannot proceed directly to mature chimpanzees.